

Lido Lend Audit Summary

Prepared by Cyfrin

Version 2.0

Lead Auditors

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Alexzoid (Formal Verification)

October 26, 2026

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1 About Cyfrin

Cyfrin is a Web3 security company dedicated to bringing industry-leading protection and education to our partners and their projects. Our goal is to create a safe, reliable, and transparent environment for everyone in Web3 and DeFi. Learn more about us at cyfrin.io.

2 About This Document

This document is a public summary of the security audit Cyfrin performed for Lido on Lido Lend. It records the repositories and commits reviewed, the timeline, auditors, in-scope files, and the number and final status of findings by severity. It closes with Cyfrin's attestation of the review's scope and final remediation status.

It is deliberately abbreviated: the technical detail of individual findings is not reproduced here. The complete audit report, version 2.0 dated October 26, 2026, was delivered to Lido, who retain it as the full record of the engagement. This summary was issued on October 29, 2026.

3 Disclaimer

The Cyfrin team makes every effort to find as many vulnerabilities in the code as possible in the given time but holds no responsibility for the findings in this document. A security audit by the team does not endorse the underlying business or product. The audit was time-boxed and the review of the code was solely on the security aspects of the in-scope code being audited. Unless the scope explicitly notes otherwise, the review is limited to the application layer and does not extend to the correctness of API/SDK/execution environment implementations, compilers, standard and third-party libraries, underlying operating systems and hardware.

4 Copyright and License

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5 Audit Overview

Over the course of 22 days, the Cyfrin team conducted an audit on the Lido Lend code provided by Lido. The findings consist of 8 Medium & 27 Low severity issues with the remainder being informational and gas optimizations.

Summary

Project Name	Lido Lend
Repositories	lido-lend-markets lido-lend lido-lend-irm
Part 1 Commits	lido-lend-markets 0e3e16af06c6... lido-lend ec7171654bdb... lido-lend-irm ce459215fd04...
Part 2 Commits	lido-lend-markets d70a17018453... lido-lend 490f9d7b3aba... lido-lend-irm 998de75d9470...
Fix Commits	lido-lend-markets bdf9e5047525... lido-lend 422d589f9d09... lido-lend-irm 566d31f7dfdf...
Public Repositories	lend 4d55f1dd98a0... lend-irm 97af4c39c51f... lend-markets 23416c43c403...
Part 1 Timeline	July 6th - July 28th, 2026
Part 2 Timeline	August 12th - August 18th, 2026
Methods	Manual Review, Formal Verification

Issues Found

Severity	Found	Resolved	Partially Resolved	Acknowledged
Critical Risk	0	0	0	0
High Risk	0	0	0	0
Medium Risk	8	4	0	4
Low Risk	27	17	0	10
Informational	21	7	1	13
Gas Optimizations	6	2	0	4
Total	62	30	1	31

6 Audit Scope

This report covers two audit phases on the same codebase. The scope for each phase was:

Part 1 (July 6th - July 28th, 2026)

```
lido-lend-markets/src/controllers/LidoLendMarketController.sol
lido-lend-markets/src/LidoLendERC4626.sol
lido-lend-markets/src/GuardianSignals.sol
lido-lend-markets/src/oracle/WstEthRateOracle.sol
lido-lend-markets/src/oracle/UsdyStableOracle.sol
lido-lend-markets/src/libs/CollateralActivationLib.sol
lido-lend-markets/src/libs/SettlementQueueLib.sol
lido-lend-markets/src/libs/ExitBookLib.sol
lido-lend-markets/src/libs/MaxHeapLib.sol
lido-lend-markets/src/libs/Constants.sol
lido-lend-markets/src/types/SettlementTypes.sol
lido-lend-markets/src/types/CollateralActivationTypes.sol
lido-lend-markets/src/types/ExitBookTypes.sol
lido-lend-markets/src/types/GuardianSignalsTypes.sol
lido-lend-markets/src/periphery/ExitBookBase.sol
lido-lend-markets/src/periphery/Erc4626ExitBook.sol
lido-lend-markets/src/periphery/LidoLendOwner.sol
lido-lend-markets/src/periphery/MorphoExitBook.sol
lido-lend-markets/src/periphery/RiskStewardDashboard.sol
lido-lend-markets/src/utils/PausableUntil.sol
lido-lend-markets/src/utils/AssetRecoverer.sol
lido-lend-markets/src/proxy/OssifiableProxy.sol
lido-lend/src/LidoLend.sol
lido-lend/src/libraries/MarketParamsLib.sol
lido-lend/src/libraries/MathLib.sol
lido-lend/src/libraries/SharesMathLib.sol
lido-lend/src/libraries/ConstantsLib.sol
lido-lend/src/libraries/ErrorsLib.sol
lido-lend/src/libraries/UtilsLib.sol
lido-lend/src/libraries/EventsLib.sol
lido-lend/src/libraries/SafeTransferLib.sol
lido-lend/src/libraries/periphery/MorphoStorageLib.sol
lido-lend/src/libraries/periphery/MorphoBalancesLib.sol
lido-lend/src/libraries/periphery/MorphoLib.sol
lido-lend-irm/src/adaptive-curve-irm/AdaptiveCurveIrm.sol
lido-lend-irm/src/adaptive-curve-irm/AdaptiveCurveConfig.sol
lido-lend-irm/src/adaptive-curve-irm/libraries/periphery/AdaptiveCurveIrmLib.sol
lido-lend-irm/src/adaptive-curve-irm/libraries/ExpLib.sol
lido-lend-irm/src/adaptive-curve-irm/libraries/MathLib.sol
lido-lend-irm/src/adaptive-curve-irm/libraries/ErrorsLib.sol
lido-lend-irm/src/adaptive-curve-irm/libraries/UtilsLib.sol
lido-lend-irm/src/common/OwnerManagerBase.sol
lido-lend-irm/src/common/OwnerManagerErrors.sol
lido-lend-irm/src/irm-router/IrmRouter.sol
lido-lend-irm/src/irm-router/libraries/ErrorsLib.sol
lido-lend-irm/src/linear-kink-irm/LinearKinkIrm.sol
lido-lend-irm/src/linear-kink-irm/libraries/ErrorsLib.sol
```

Part 2 (August 12th - August 18th, 2026)

```
lido-lend-markets/src/controllers/LidoLendMarketController.sol
lido-lend-markets/src/LidoLendERC4626.sol
lido-lend-markets/src/GuardianSignals.sol
lido-lend-markets/src/adapters/EthereumGeneralAdapter1.sol
lido-lend-markets/src/adapters/LidoLendVaultAdapter.sol
lido-lend-markets/src/libs/CollateralActivationLib.sol
lido-lend-markets/src/libs/ExitBookLib.sol
lido-lend-markets/src/libs/MaxHeapLib.sol
lido-lend-markets/src/libs/SettlementQueueLib.sol
lido-lend-markets/src/types/ExitBookTypes.sol
lido-lend-markets/src/types/GuardianSignalsTypes.sol
lido-lend-markets/src/types/SettlementTypes.sol
lido-lend-markets/src/periphery/ExitBookBase.sol
lido-lend-markets/src/periphery/Erc4626ExitBook.sol
lido-lend-markets/src/periphery/MorphoExitBook.sol
lido-lend-markets/src/periphery/LidoLendMarketFactory.sol
lido-lend-markets/src/periphery/LidoLendOwner.sol
lido-lend-markets/src/periphery/RiskStewardDashboard.sol
lido-lend-markets/src/utils/AssetRecoverer.sol
lido-lend/src/LidoLend.sol
lido-lend/src/libraries/ErrorsLib.sol
lido-lend-irm/src/adaptive-curve-irm/AdaptiveCurveIrm.sol
lido-lend-irm/src/adaptive-curve-irm/libraries/AdaptiveCurveMathLib.sol
lido-lend-irm/src/adaptive-curve-irm/libraries/periphery/AdaptiveCurveIrmLib.sol
lido-lend-irm/src/irm-config-factory/IrmConfigFactory.sol
lido-lend-irm/src/linear-kink-irm/LinearKinkConfig.sol
lido-lend-irm/src/linear-kink-irm/LinearKinkIrm.sol
```

The repositories `lido-lend`, `lido-lend-irm` and `lido-lend-markets` are now published as `lend`, `lend-irm` and `lend-markets`. The public commits in the Summary table hold the audited contracts with all fixes applied.

7 In-Scope File Checksums

These keccak256 digests cover the exact bytes of every in-scope source file at the final audited state, with all fixes applied. A single changed byte changes the digest.

Public repository verification (October 26, 2026). The repositories `lido-lend`, `lido-lend-irm` and `lido-lend-markets` are now published as `lend`, `lend-irm` and `lend-markets`, respectively. Cyfrin independently recomputed all 53 file hashes at the following public commits and confirmed that every digest matches:

- lidofinance/lend at `4d55f1dd98a0f7712543965c66e03c963a004263`
- lidofinance/lend-irm at `97af4c39c51f1e023d8025319f34a1658036066c`
- lidofinance/lend-markets at `23416c43c4038ea3c00a13d6203949c8f88905cc`

Files use `src/...` in each repository; `lend-markets` dependencies are under `lib/lend/` and `lib/lend-irm/`. The aggregate below retains the original paths: `src/...` for markets, `lib/lido-lend/...` and `lib/lido-lend-irm/...`.

lido-lend-markets (25 files)

```
src/GuardianSignals.sol
  0x394c7c7420c828717a5d644442ac2ae9ea3e6c37cb3b0afc6a765f5d39da7f28
src/LidoLendERC4626.sol
  0xaf2223efeb83069af02774b0b9c64c1eb950fd909b1a7b34345b0a798b8fe8d6
src/adapters/EthereumGeneralAdapter1.sol
  0xa21c89addca6293d18be340a4255b721cc098c144bdbc8ff83d5d62f839a43d9
src/adapters/LidoLendVaultAdapter.sol
  0x49ab46019a882ad147564534717a03d67976db41a1828b812e7da0f53bb8f5f9
src/controllers/LidoLendMarketController.sol
  0x373eaab44d8ebe4711a9c5fee5f5030aca5475c2805d02f1075a1e6e174d8f0f
```

```
src/libs/CollateralActivationLib.sol
  0xd8c2acee361f6fd8a907825c2332ec80471db9e46925f8f6b3806a2ac6880e75
src/libs/Constants.sol
  0x2898cb8a036aef5d7d66501926b6d028cf29bdc47baad01f037955d7380050a0
src/libs/ExitBookLib.sol
  0x2ccdcdfdb9b8949b05b85f0c6aa748eb75c48e87456f040ffdc4d365ea8b38f0
src/libs/MaxHeapLib.sol
  0x268bbf3e5a8c40aa38933faa4d5bed7f3a00dda0dda936f5eb04dda29e41763e
src/libs/SettlementQueueLib.sol
  0x5acbdaf61af72bbc283cb4b2470a24c84f6423a10d207bb85dc45b409b167147
src/oracle/UsdyStableOracle.sol
  0x82caeb8b1e50738dc979cd9469a9da63f34d86c15010d4934aefd767495a2825
src/oracle/WstEthRateOracle.sol
  0xa7eb3dca147913db68745c9b66c7e28a42a93f0ca19e2390fed69e0d6c451114
src/periphery/Erc4626ExitBook.sol
  0xa85f0a11fa3625a81ffa32b64af5c8a964ea23cc7190bf009472333bba643c14
src/periphery/ExitBookBase.sol
  0x412160d3b187b34552ab8cd47b17dea4b569e32b6e88c16c255acd44ad937d97
src/periphery/LidoLendMarketFactory.sol
  0xe680a0610bd7d43ada1406e1387fe9aa1a741c03835273dbeac6eb90d55104c5
src/periphery/LidoLendOwner.sol
  0xcdf03ccaef93ea6b6a6a57e6800a955ba9ca0bfe13918b14238bbf0086f9db72
src/periphery/MorphoExitBook.sol
  0xf0da6a947793953ce0cd4135716c60ce81bca88a8aab8c845a2db89f71917e21
src/periphery/RiskStewardDashboard.sol
  0x90d7d44e7e62868693e4c4f4ce4322ae9631a875b4b2f876685fac6fda6d2e9c
src/proxy/OssifiableProxy.sol
  0x4cd5b76eac5248fa2e7631eb081094e21afbcdb84fe39d195120d35eaa6bdebe
src/types/CollateralActivationTypes.sol
  0xeeeb1486296e0adc4d4844665cee1fc393713d9b6efa5ea801d081bc32603c7b
src/types/ExitBookTypes.sol
  0xfd114855a04301d306f62d98096fa8d38730c339180b6c05eb8048c5f93d3d8b
src/types/GuardianSignalsTypes.sol
  0xf7cb4606fa6355ecd162a9f1355b4abd3ba19210189a96bfe5c25a62a8dc3f15
src/types/SettlementTypes.sol
  0xb11bc798eda3262e576c61ee69abdbb43ca672a4a96a4930eb43031363347849
src/utils/AssetRecoverer.sol
  0x5ad79a83e23c26dcbd3c5c64d090d8faf05898cc5eb34d6faaecfa960f075bea
src/utils/PausableUntil.sol
  0xc4f4a94353ff81a7351d1d7758ab63cd179ad70e031fa6244a9b9696ce0f0ca0
```

lido-lend (12 files)

```
src/LidoLend.sol
  0xc9f69ab05c6672a1571b4b91d0ad99b27e87bc64c6296c67a73c41a8071de1df
src/libraries/ConstantsLib.sol
  0x34f33fef6867defa982f42d0d543ec23d62a17580ba628dd109ad3a547e58637
src/libraries/ErrorsLib.sol
  0x9d9ca277430b17d8781dafa12a0f3f02b0d8401a343eb0ec380d8d9ae34a359d
src/libraries/EventsLib.sol
  0x7a8f5e2416ba393f5113a43e051cc5257c1975cb47171babc78d143dd4ffbde8
src/libraries/MarketParamsLib.sol
  0x6c8e25952be253b960283e8caecf8fe3138e1b2ab0595e67055d56a3eeabeffc
src/libraries/MathLib.sol
  0xa7354cbbcecef7bc0c94b61061c4e5da75515056b8e2db65e826b00d7369744a
src/libraries/SafeTransferLib.sol
  0x120117a636a3a46d6ba7e30ba00ece4e6c284fc25d7d990a841f4abaced430b1
src/libraries/SharesMathLib.sol
  0xe6d1c0ee97239bcda3ca8429fdbad2486faf4192312ce6c4b322772e038f01a8
src/libraries/UtilsLib.sol
  0x9a98b6d9cd11924d5a15906046b7c26816a74c61abf9a3d51240a11a78983276
src/libraries/periphery/MorphoBalancesLib.sol
  0x37dd5f303f0d36a211fdf6da9ad98d7f4efeaf6f81183bcf20bf4202d4515f26
src/libraries/periphery/MorphoLib.sol
  0x7d59323645f4977e73c388cdb3aa7aeb393fc18251e9249e30ad6ece3b92908a
src/libraries/periphery/MorphoStorageLib.sol
  0x5d2b8dbbdf1c7aa09202b8026d548cdd4ef19a60e0f400f709f76e061d1fd4c1
```

lido-lend-irm (16 files)

```
src/adaptive-curve-irm/AdaptiveCurveConfig.sol
  0x06ee168c37530dfb8ef6cdb16f67b8234d709624f1698f232126307dcdf16aef
src/adaptive-curve-irm/AdaptiveCurveIrm.sol
  0xe289c4166783038c6c9304a5f882b38860a04a4f01682f9801e588876b1020af
src/adaptive-curve-irm/libraries/AdaptiveCurveMathLib.sol
  0x441ef1a01e6aa1db49b56207779037b8f86c794d804e64eaf31f7f5a2af4a44e
src/adaptive-curve-irm/libraries/ErrorsLib.sol
  0xb1c732964472582571c74884e9903e102cf9e531b1429699803e08eeb2b994a9
src/adaptive-curve-irm/libraries/ExpLib.sol
  0x32e269f72ad4f1938f8c78e4ecfce741ede5358b08897154d8b5662307a8aa49
src/adaptive-curve-irm/libraries/MathLib.sol
  0x0cde8ff301e1544641526b64fc48316cb2c37d6eb5e3b067444d4a1c4aa52aed
src/adaptive-curve-irm/libraries/UtilsLib.sol
  0xd4b7d2e137bd50fd5176213b5ae9af6e7dc3fcc305ccd005734bb0158594ba4d
src/adaptive-curve-irm/libraries/periphery/AdaptiveCurveIrmLib.sol
  0xf44fcf95334f381705d105bc4c3a2d3d136c4874cc400e7a884c09b10226c5b5
src/common/OwnerManagerBase.sol
  0x4bf1527d81d47d4732fb40ab4bbe192d551979df63c43bb8cc47d45ae70115cb
src/common/OwnerManagerErrors.sol
  0xd54420aecdcdbd3102ebeb5fb897e8719899f8ac716ffdc0ae4e9b8d5b3b7bf5
src/irm-config-factory/IrmConfigFactory.sol
  0x4f643c3d9c111fd54b6e655f695fdbf50284a5d7bb6a64a440c3ae27d8fb81c4
src/irm-router/IrmRouter.sol
  0x1f0a8a118827fe8d16434cf20505bdc4898b64a2835371eb426fbdb845d43beb
src/irm-router/libraries/ErrorsLib.sol
  0x700ea721dceac328b83ea7ac772d0c8af8fc76bf329be9334e03162cf04a7a4a
src/linear-kink-irm/LinearKinkConfig.sol
  0x39f8ec370a84976346fb2ad3b5acb2bc9d9c6c598332a486674a428547a33ae7
src/linear-kink-irm/LinearKinkIrm.sol
  0x23385aec7fcb622627671871b2035c05cf49cb9451ab60dfe90edf7c5a262e6e
src/linear-kink-irm/libraries/ErrorsLib.sol
  0x7e15cb72d5068b3889b98ec1ad4929548dcbcd10fc0c0da70d747125d07cfbbe
```

Aggregate root: keccak256 of rows sorted by original path, each digest + two spaces + path, joined by newlines without a trailing newline:

```
0x84be32b38acdf912b6ae7fbaabe235398f03d682394d121dcdc41af042b4ecc4
```

8 Risk Classification

	Impact: High	**Impact: Medium**	**Impact: Low**
Likelihood: High	Critical	High	Medium
Likelihood: Medium	High	Medium	Low
Likelihood: Low	Medium	Low	Low

9 Protocol Summary

Lido Lend is an over-collateralized lending protocol that pairs a Morpho Blue fork core with Lido-specific extensions. The core (LidoLend) keeps Morpho's isolated-market supply, borrow, collateral, liquidation, and flash-loan accounting, and adds a settle primitive that cancels a borrower's debt against a lender's supply shares in exchange for collateral, and hooks that route every core action on a controlled market to its LidoLendMarketController.

The controller and its periphery add:

- a single-market ERC-4626 vault (LidoLendERC4626) that is the market's sole lender and exposes a settle-based exit
- a settlement queue that lets lenders of an illiquid market exit by settling borrowers ahead of natural repayment
- an exit book where a lender posts an exit request that any filler closes from available liquidity for a Dutch-auction priority fee
- a collateral-activation throttle that delays newly supplied collateral from becoming borrowable
- a per-market guardian-signal system
- a market factory (LidoLendMarketFactory) that deploys and wires each market's controller, vault, and guardian registry
- bundler adapters (EthereumGeneralAdapter1, LidoLendVaultAdapter) that batch native-ETH staking, wstETH wrapping, and vault entry into a single transaction
- oracle adapters pricing wstETH (WstEthRateOracle) and USDY (UsdyStableOracle) as collateral

Interest is priced by an adaptive-curve model or a linear-kink model selected per market through an IRM router. The launch market borrows WETH against Lido wstETH collateral on Ethereum mainnet.

10 Attestation

Cyfrin attests that it performed a security review of the Lido Lend code provided by Lido.

The repositories, commits, review period and findings with their final status are in the Summary and Issues Found tables in Section 5, and the in-scope files in Section 6. This attestation is limited to the documented scope, audited commits and mitigations specifically reviewed by Cyfrin.

Cyfrin publishes this document and its SHA-256 hash at https://github.com/Cyfrin/cyfrin-audit-reports. To check a copy on macOS or Linux, run the following in the folder that contains it; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name. Before running it, check that the URL is Cyfrin's repository.

```
U=https://github.com/Cyfrin/cyfrin-audit-reports/raw/main/reports/sha256
N=2026-10-26-cyfrin-lido-lend-v2.0.pdf
F=$N
H=$(curl -fsSL $U/$N.sha256 | cut -d' ' -f1)
echo "$H *$F" | shasum -a 256 -c
```

Cyfrin's full audit report with all findings provided to Lido has the SHA-256 hash and file name below. To check a copy of it, first check this document as above, then run the following in the folder that contains the full report; it prints `OK` when the copy matches. If the copy was renamed, set F to its file name.

```
H=8143fa9e14f2e925f9a858b75db428e689b81eca4a6954523e15fc51fc68f6db
F=2026-10-26-cyfrin-lido-lend-v2.0-full.pdf
echo "$H *$F" | shasum -a 256 -c
```

On Linux, `sha256sum -c` can be used in place of `shasum -a 256 -c`. On Windows, run `Get-FileHash <file> -Algorithm SHA256` in PowerShell and compare the result with the hash.